

BUTLERS GORGE, AustraliaWMO: **949590**Lat: **42.2753S**Lon: **146.2758E**Elev: **667**StdP: **93.56**Time Zone: **10.00 (AUS)**Period: **96-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	-3.7	-2.9	-6.6	2.3	2.0	-4.9	2.7	-0.2	12.5	2.5	11.1	1.0	1.7	240	0.641

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	11.9	26.0	14.9	23.7	14.3	21.7	13.7	16.5	22.7	15.5	21.3	14.5	19.9	3.8	310

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.5	11.2	18.0	13.3	10.3	17.0	12.2	9.6	16.3	48.6	22.9	45.5	21.4	42.9	20.1	21.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11.1	9.4	7.8	-5.7	30.1	0.8	2.7	-6.2	32.1	-6.7	33.6	-7.1	35.2	-7.6	37.1
			-6.1	18.5	0.6	1.3	-6.6	19.4	-6.9	20.2	-7.3	20.9	-7.7	21.8
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	7.6	12.6	12.3	10.7	7.7	5.6	4.3	3.4	3.8	5.2	6.7	8.7	10.3
	DBStd	4.43	3.72	3.67	3.74	3.00	2.58	2.18	2.23	2.24	2.68	3.27	3.84	3.84
	HDD10.0	1224	15	17	38	79	137	170	204	193	145	112	69	44
	HDD18.3	3933	183	172	239	318	394	420	463	452	393	360	289	251
	CDD10.0	340	94	81	58	11	2	0	0	0	2	10	31	52
	CDD18.3	8	4	2	1	0	0	0	0	0	0	0	0	1
	CDH23.3	240	118	41	23	0	0	0	0	0	0	3	17	38
	CDH26.7	50	32	6	4	0	0	0	0	0	0	0	2	6
(14) Wind	WSAvg	3.0	2.9	3.2	2.8	2.9	3.1	2.4	2.8	2.8	3.0	3.3	3.2	3.4
(15) Precipitation	PrecAvg	1649	99	76	95	146	158	142	175	172	168	160	137	126
	PrecMax	2282	224	186	215	345	427	405	389	331	452	567	384	267
	PrecMin	1088	33	18	27	51	39	56	76	42	38	42	39	6
	PrecStd	290	47	44	44	66	84	70	76	80	77	88	64	62
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	30.1	27.3	26.3	19.5	15.2	11.7	10.6	13.1	17.0	22.6	25.7	27.2
		MCWB	16.7	16.1	15.7	12.9	10.0	9.1	7.6	8.6	10.5	12.8	14.1	14.6
	2%	DB	26.6	24.2	22.4	17.0	12.7	10.5	9.2	11.0	13.9	18.6	22.2	23.7
		MCWB	15.3	15.1	14.6	11.5	9.4	8.1	6.9	7.4	8.9	10.5	13.1	13.6
	5%	DB	23.6	22.2	20.1	15.1	11.4	9.4	8.4	9.3	12.0	16.0	19.4	20.7
		MCWB	14.3	14.3	13.9	11.0	9.0	7.5	6.3	6.7	8.0	10.1	11.9	12.8
	10%	DB	21.0	19.9	17.7	13.3	10.0	8.5	7.5	8.0	10.3	13.7	16.4	18.2
		MCWB	13.5	13.6	13.0	10.1	8.0	6.8	5.7	6.0	7.3	9.2	10.9	12.3
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	18.8	17.5	16.9	14.1	11.1	9.8	8.6	9.1	10.9	13.6	15.3	16.7
		MCDB	26.3	23.1	23.2	18.2	13.3	11.0	9.8	11.8	15.9	17.9	22.1	22.2
	2%	WB	16.5	16.2	15.6	12.6	9.9	8.6	7.2	8.0	9.4	11.9	14.2	14.9
		MCDB	23.9	21.5	20.6	15.8	12.0	9.9	8.8	10.5	12.7	17.1	20.3	20.6
	5%	WB	15.1	15.3	14.5	11.4	9.1	7.7	6.5	7.0	8.4	10.5	12.9	13.7
		MCDB	21.9	20.2	18.8	14.3	10.9	9.1	8.1	9.0	11.2	15.2	18.0	19.4
	10%	WB	13.9	14.1	13.3	10.3	8.2	6.9	5.8	6.1	7.5	9.1	11.4	12.5
		MCDB	19.8	18.8	17.5	13.1	10.0	8.4	7.4	7.8	10.1	13.6	16.2	18.0
(35) Mean Daily Temperature Range	5% DB	MDBR	11.9	11.5	10.1	8.7	6.7	6.6	6.2	6.8	7.9	9.9	10.4	10.8
		MCDBR	18.4	16.0	15.5	13.1	9.9	8.7	8.1	10.1	11.9	15.5	16.8	17.1
	5% WB	MCWBR	8.5	8.0	8.2	8.1	6.9	6.8	6.1	6.9	7.3	8.9	8.6	8.5
		MCWBR	15.9	13.6	13.5	11.7	8.7	7.5	7.3	9.2	10.2	14.1	15.0	14.5
(40) Clear-Sky Solar Irradiance	taub	taub	0.312	0.305	0.301	0.292	0.279	0.269	0.271	0.290	0.307	0.309	0.307	0.306
		taud	2.587	2.624	2.624	2.626	2.642	2.653	2.653	2.594	2.534	2.545	2.579	2.597
	Ebn at Noon	Ebn at Noon	1016	998	958	898	844	821	846	886	933	981	1015	1027
		Edh at Noon	102	93	85	72	60	54	58	73	91	100	102	102
(44) All-Sky Solar Radiation	RadAvg	6.48	5.61	4.13	2.79	1.69	1.34	1.50	2.24	3.32	4.78	5.88	6.34	6.44
	RadStd	0.49	0.35	0.24	0.18	0.13	0.12	0.09	0.18	0.22	0.31	0.47	0.44	0.44

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (34 neighbors)	+0.32	N/A	N/A	+0.67	+0.33	N/A	N/A	-98	+106	+12

Nomenclature: See separate page